SPA617: RF Components Characterization – Filter, Mixer, and Power Splitter/Combiner

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1 Aim of the Experiment

To understand the operation, underlying transmission/reflection theories, and experimentally characterize the passive and active frequency translation components of an RF receiver chain, specifically:

1. Characterizing an RF filter to identify its type, passband/stopband response, insertion loss, cutoff frequencies, input reflection coefficient ($|\Gamma|$), and return loss (RL).
2. Demonstrating a quarter-wave open-circuit coaxial stub acting as a notch filter and comparing calculated versus measured notch frequencies.
3. Observing frequency down-conversion/up-conversion in a three-port nonlinear RF mixer and evaluating its conversion loss.
4. Measuring the power division, path balance, and insertion loss of a 2-way power splitter/combiner.

2 Theory and Principles

2.1 RF Filters and Transmission Response

RF filters select desired frequency bands and attenuate unwanted signals or interference in a receiver front-end. The primary response categories are:

- **Low-Pass Filter (LPF):** Passes $f < f_c$ and attenuates $f > f_s$.
- **High-Pass Filter (HPF):** Passes $f > f_c$ and attenuates $f < f_s$.
- **Band-Pass Filter (BPF):** Passes $f_1 < f < f_2$ with center frequency $f_0 = \sqrt{f_1 f_2}$.
- **Notch/Band-Reject Filter:** Attenuates $f_1 < f < f_2$ around notch frequency f_0.

The transmission behavior is quantified by the **Insertion Loss (IL)**:

$$\text{IL (dB)} = P_{\text{in}} \text{ (dBm)} - P_{\text{out}} \text{ (dBm)} \tag{1}$$

where P_{in} is the reference power measured directly without the filter (evaluated at $P_{\text{in}} = -0.9$ dBm taking into account cable attenuation from the generator), and P_{out} is the power measured after inserting the filter into the signal path. The 3 dB cutoff frequency f_c is defined where the insertion loss drops by 3 dB relative to the passband baseline.

2.2 Filter Impedance Match, Reflection, and Return Loss

When a filter's input impedance Z_{in} differs from the system characteristic impedance $Z_0 = 50\,\Omega$, a fraction of the incident wave is reflected back to the source. The complex voltage reflection coefficient Γ is given by:

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0} \tag{2}$$

Using a 3-port Tee-connector with a $1\,\text{M}\Omega$ high-impedance oscilloscope probe, the standing wave envelope formed by incident (V_{inc}) and reflected (V_{ref}) waves yields maximum (V_{max}) and minimum (V_{min}) node voltages:

$$V_{\text{inc}} = \frac{V_{\text{max}} + V_{\text{min}}}{2}, \qquad V_{\text{ref}} = \frac{V_{\text{max}} - V_{\text{min}}}{2} \tag{3}$$

The magnitude of the reflection coefficient $|\Gamma|$ and the **Return Loss (RL)** are:

$$|\Gamma| = \frac{V_{\text{max}} - V_{\text{min}}}{V_{\text{max}} + V_{\text{min}}} \tag{4}$$

$$\text{RL (dB)} = -20 \log_{10} |\Gamma| \tag{5}$$

A well-matched filter exhibits $|\Gamma| \to 0$ (RL $\gg 10$ dB), while full mismatch yields $|\Gamma| = 1$ (RL $= 0$ dB).

2.3 Quarter-Wave Open-Circuit Stub Notch Filter

An open-ended coaxial line of physical length L connected in parallel (shunt) across an RF line behaves as a resonant stub transformer. At its quarter-wave fundamental resonance, the open circuit ($Z = \infty$) at the far end is transformed into an approximate short-circuit ($Z \approx 0$) at the Tee junction, creating a deep rejection notch.

The theoretical notch frequency f_0 is given by:

$$f_0 = \frac{\text{VF} \cdot c}{4L} \tag{6}$$

where VF is the cable velocity factor (≈ 0.66 for standard RG-58 coaxial cable), $c = 3 \times 10^8$ m/s, and L is the physical length of the stub.

2.4 RF Mixer Operation and Conversion Loss

An RF mixer is a three-port nonlinear device (RF, LO, and IF ports). Applying an RF signal at frequency f_{RF} and a Local Oscillator drive at f_{LO} generates sum and difference intermodulation frequencies:

$$f_{\text{IF}} = f_{\text{RF}} \pm f_{\text{LO}} \tag{7}$$

In down-conversion, the difference frequency $f_{\text{IF}} = |f_{\text{RF}} - f_{\text{LO}}|$ is selected, while up-conversion utilizes $f_{\text{IF}} = f_{\text{RF}} + f_{\text{LO}}$. The efficiency of this process is measured by the **Conversion Loss (CL)**:

$$\text{Conversion Loss (dB)} = P_{\text{RF}}\ \text{(dBm)} - P_{\text{IF}}\ \text{(dBm)} \tag{8}$$

2.5 Power Splitter / Combiner

A 2-way equal power splitter divides input signal power P_{in} equally between two output ports. For an ideal loss-free splitter, each output port receives half the power (a 3 dB division drop). Practical resistive/reactive splitters introduce additional internal dissipation loss.

The insertion losses to output ports 1 and 2 are:

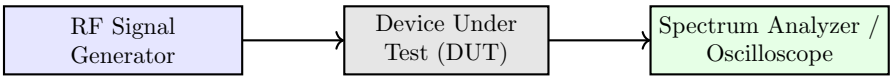
$$\text{Loss}_1\ \text{(dB)} = P_{\text{in}}\ \text{(dBm)} - P_{\text{out1}}\ \text{(dBm)} \tag{9}$$

$$\text{Loss}_2\ \text{(dB)} = P_{\text{in}}\ \text{(dBm)} - P_{\text{out2}}\ \text{(dBm)} \tag{10}$$

In power combiner mode, coherent signals at frequencies f_1 and f_2 combine into a single output port. For identical coherent inputs, constructive interference doubles the voltage (+6 dB amplitude addition relative to individual voltage paths), yielding net output amplification minus internal insertion loss.

3 Experimental Setups

A) Transmission Measurement Setup:



B) Reflection / Return Loss Measurement Setup:

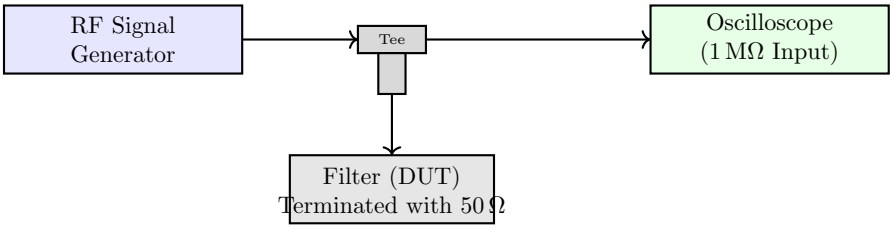


Figure 1: Experimental arrangements for transmission characterization (A) and filter return-loss estimation (B).

4 Observation Tables and Quantitative Analysis

4.1 Part A1: RF Filter Transmission Response and Insertion Loss

Reference Input Power (P_{in}): −0.90 dBm (measured at cable end on Spectrum Analyzer)

Table 1: Transmission response data for RF filter characterization

Frequency (MHz)	**Reference P_{in} (dBm)**	**Filter P_{out} (dBm)**	**Insertion Loss IL (dB)**
100	-0.90	-1.36	0.46
150	-0.90	-1.59	0.69
200	-0.90	-1.84	0.94
250	-0.90	-1.70	0.80
300	-0.90	-1.89	0.99
350	-0.90	-2.50	1.60
400	-0.90	-2.22	1.32
450	-0.90	-3.09	2.19
500	-0.90	-3.26	2.36
550	-0.90	-4.72	3.82
600	-0.90	-8.79	7.89
650	-0.90	-21.92	21.02
700	-0.90	-28.95	28.05

4.2 Part A2: Filter Input Reflection Coefficient and Return Loss

Selected Frequency Range in Passband: 61–70 MHz

Table 2: Tee-junction voltage envelope and calculated return loss

Frequency (MHz)	V_{max} (mV)	V_{min} (mV)	V_{inc} (mV)	V_{ref} (mV)	$\lvert\Gamma\rvert$	**Return Loss RL (dB)**
61–70	136.0	132.0	134.0	2.0	0.0149	**36.52**

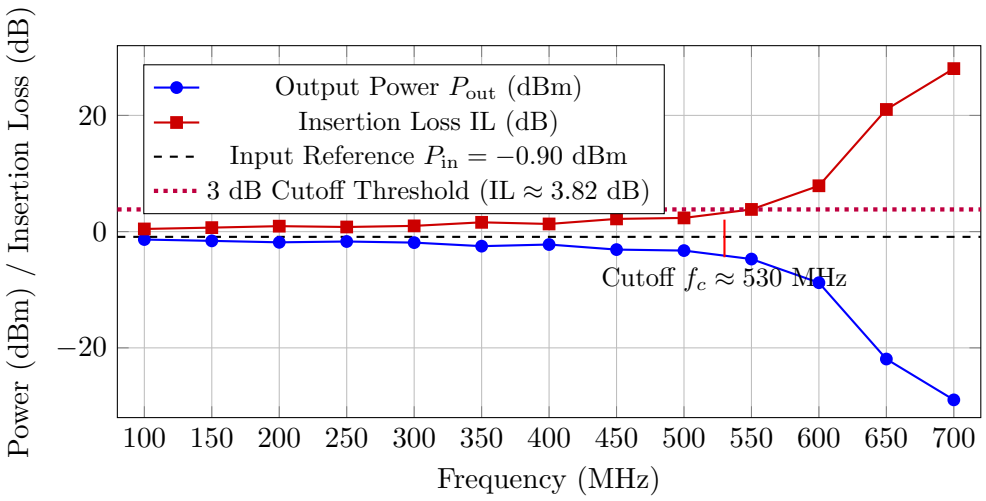


Figure 2: Measured Filter Output Power and Insertion Loss vs. Frequency identifying a **Low-Pass Filter** with $f_c \approx 530$ MHz.

Calculation:

$$V_{\text{inc}} = \frac{136.0 + 132.0}{2} = 134.0 \text{ mV}, \qquad V_{\text{ref}} = \frac{136.0 - 132.0}{2} = 2.0 \text{ mV}$$

$$|\Gamma| = \frac{2.0}{134.0} = \mathbf{0.01493}$$

$$\text{RL} = -20 \log_{10}(0.01493) = \mathbf{36.52} \text{ dB}$$

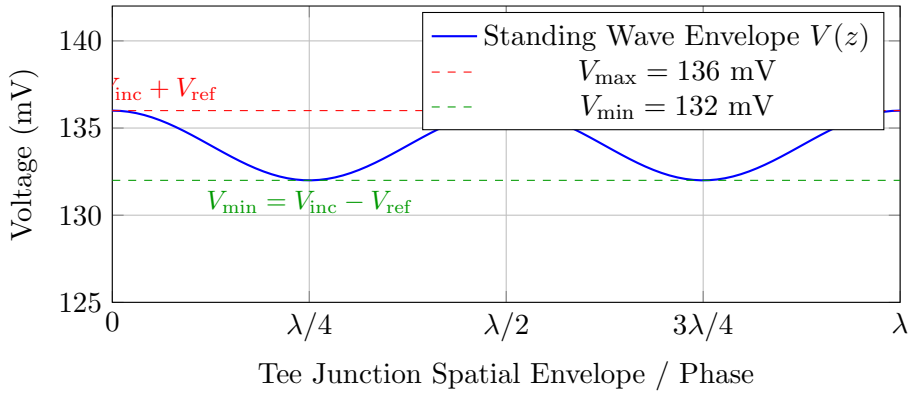


Figure 3: Visualization of Tee-junction Standing Wave Voltage Envelope ($V_{\text{max}} = 136$ mV, $V_{\text{min}} = 132$ mV), illustrating high transmission match ($|\Gamma| = 0.0149$, RL = 36.52 dB).

4.3 Part A3: Quarter-Wave Open-Circuit Stub Notch Filter

Measured Stub Physical Length (L): 0.91 m **Assumed Velocity Factor (VF):** 0.66
Calculated Theoretical Notch Frequency ($f_{0,\mathbf{calc}}$):

$$f_{0,\text{calc}} = \frac{0.66 \times 3 \times 10^8}{4 \times 0.91} = 54.395 \text{ MHz} \approx \mathbf{54.40} \text{ MHz} \tag{11}$$

Table 3: Transmission measurement for quarter-wave open stub filter

Frequency (MHz)	**Reference P_{in} (dBm)**	**Output P_{out} (dBm)**	**Attenuation (dB)**
50	-0.90	-9.20	8.30
51	-0.90	-10.30	9.40
52	-0.90	-11.40	10.50
53	-0.90	-13.20	12.30
54	-0.90	-15.40	14.50
55	-0.90	-29.40	**28.50**
56	-0.90	-22.60	21.70
57	-0.90	-27.80	26.90
58	-0.90	-25.40	24.50
59	-0.90	-20.60	19.70
60	-0.90	-17.50	16.60
61	-0.90	-15.10	14.20

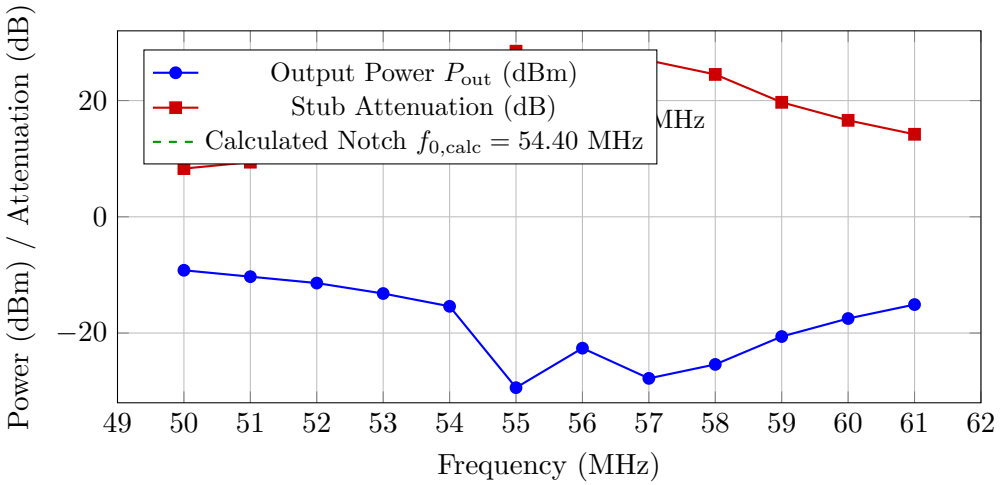


Figure 4: Output power and rejection attenuation profile of the quarter-wave open stub showing maximum notch depth at 55.0 MHz.

4.4 Part B: RF Mixer Down/Up-Conversion Characterization

LO Frequency ($f_{\mathbf{LO}}$): 10.0 MHz **LO Drive Power ($P_{\mathbf{LO}}$):** +1.04 dBm **RF Input Power ($P_{\mathbf{RF}}$):** −0.90 dBm

Table 4: Mixer frequency translation and conversion loss

f_{RF} (MHz)	Mixed Sideband	Expected f_{IF} (MHz)	Measured f_{IF} (MHz)	P_{IF} (dBm)	**Conversion Loss**
13	Up ($f_{\mathrm{RF}} + f_{\mathrm{LO}}$)	23.0	23.0	-13.80	**12.90**
13	Down ($f_{\mathrm{RF}} - f_{\mathrm{LO}}$)	3.0	3.0	-15.00	**14.10**
23	Up ($f_{\mathrm{RF}} + f_{\mathrm{LO}}$)	33.0	33.0	-14.10	**13.20**
23	Down ($f_{\mathrm{RF}} - f_{\mathrm{LO}}$)	13.0	13.0	-13.80	**12.90**
33	Up ($f_{\mathrm{RF}} + f_{\mathrm{LO}}$)	43.0	43.0	-10.08	**9.18**
33	Down ($f_{\mathrm{RF}} - f_{\mathrm{LO}}$)	23.0	23.0	-10.40	**9.50**
43	Up ($f_{\mathrm{RF}} + f_{\mathrm{LO}}$)	53.0	53.0	-10.57	**9.67**
43	Down ($f_{\mathrm{RF}} - f_{\mathrm{LO}}$)	33.0	33.0	-10.61	**9.71**

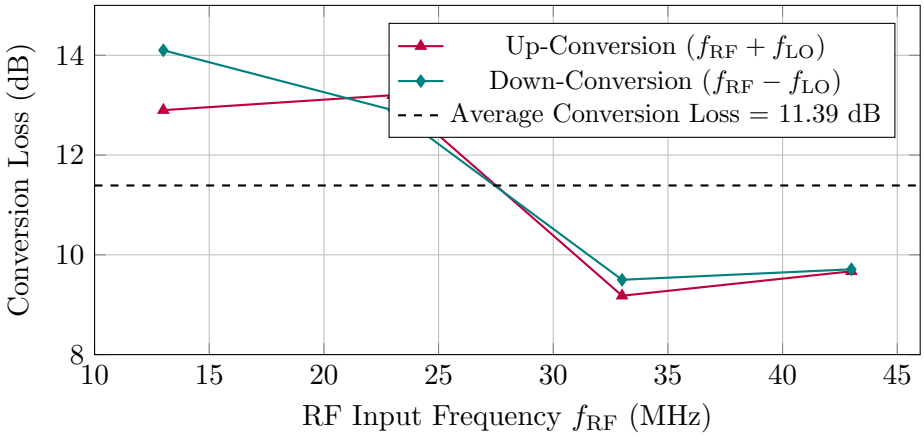


Figure 5: Mixer Conversion Loss vs. Input RF Frequency for Up-conversion and Down-conversion sidebands.

4.5 Part C: Power Splitter / Combiner Characterization

Reference Input Power (P_{in}): -0.90 dBm

4.5.1 2-Way Power Splitter Mode

Table 5: 2-Way Power Splitter path power division and insertion loss

Frequency (MHz)	P_{in} (dBm)	P_{out1} (dBm)	P_{out2} (dBm)	**Loss$_1$ (dB)**	**Loss$_2$ (dB)**
33	-0.90	-4.42	-4.40	**3.52**	**3.50**
43	-0.90	-4.46	-4.44	**3.56**	**3.54**
53	-0.90	-4.47	-4.43	**3.57**	**3.53**

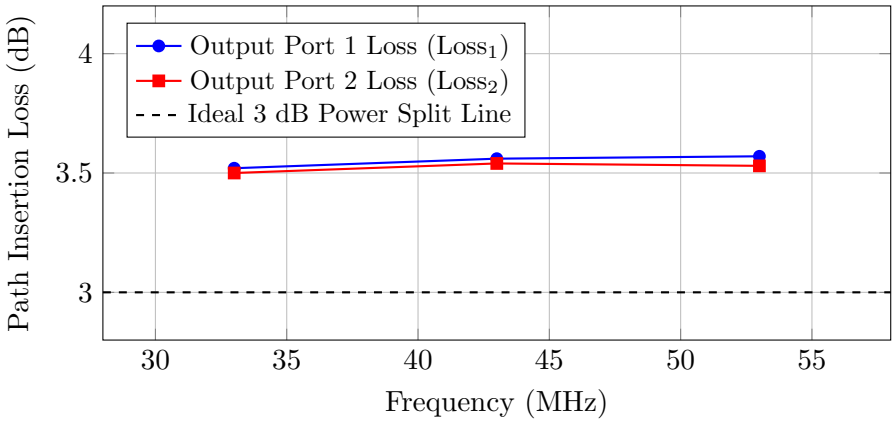


Figure 6: Power Splitter Path Losses ($\text{Loss}_1, \text{Loss}_2$) compared against ideal 3.0 dB division.

4.5.2 Power Combiner Mode

- **Input 1 ($f_1 = 10$ MHz):** $P_{\text{in1}} = -0.90$ dBm ($V_1 = 201.8$ mV$_{\text{peak}}$ into 50 Ω)
- **Input 2 ($f_2 = 10$ MHz):** $P_{\text{in2}} = +1.64$ dBm ($V_2 = 270.4$ mV$_{\text{peak}}$ into 50 Ω)
- **Combined Output Power Measured (P_{comb}):** $+3.05$ dBm

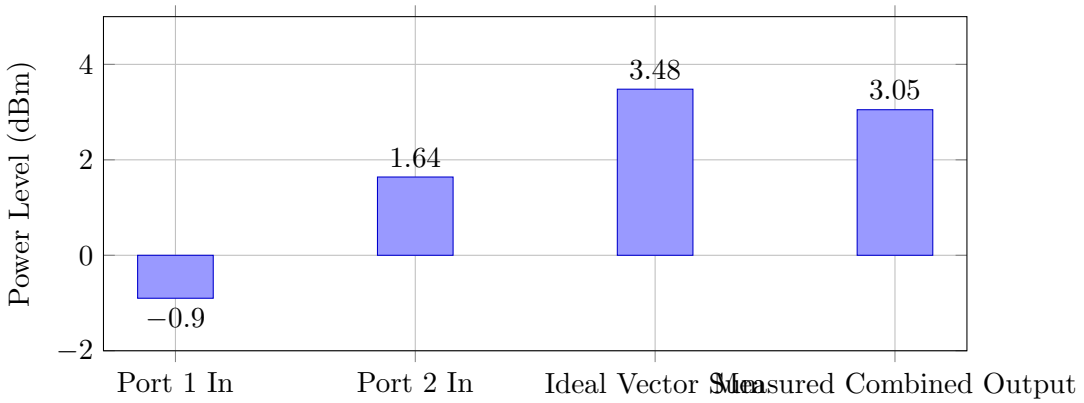


Figure 7: Power Combiner operation showing coherent signal summation ($P_1 = -0.90$ dBm, $P_2 = +1.64$ dBm yielding $+3.05$ dBm output).

5 Summary of Experimental Results

Table 6: Principal Measured and Derived Parameters

Parameter / Quantity	**Symbol**	**Measured / Calculated Value**
Filter Type Identified	—	Low-Pass Filter (LPF)
Cutoff Frequency (3 dB)	f_c	≈ 530 MHz
Minimum Passband Insertion Loss	$\mathrm{IL}_{\min}$	0.46 dB (at 100 MHz)
Filter Reflection Coefficient in Passband	$\lvert\Gamma\rvert$	0.0149
Filter Passband Return Loss	RL	36.52 dB
Calculated Stub Notch Frequency	$f_{0,\mathrm{calc}}$	54.40 MHz
Measured Stub Notch Frequency	$f_{0,\mathrm{meas}}$	55.00 MHz
Average Mixer Conversion Loss	$\mathrm{CL}_{\mathrm{avg}}$	11.39 dB
Splitter Output Division Loss per Path	$\mathrm{Loss}_1, \mathrm{Loss}_2$	≈ 3.55 dB, 3.52 dB (≈ 0.54 dB excess)
Power Combiner Output (10 MHz coherent)	P_{comb}	+3.05 dBm (0.43 dB excess loss)

6 Conclusions and Discussion

In this experiment, the passive and active RF components forming an RF receiver front-end were characterized:

1. **RF Filter Characterization:** The Device Under Test was identified as a **Low-Pass Filter** with a 3 dB cutoff frequency $f_c \approx 530$ MHz. In the passband (100–400 MHz), insertion loss remained < 1.5 dB, while rising sharply to > 28 dB at 700 MHz. The Tee-connector reflection measurement yielded $|\Gamma| = 0.0149$ and RL = 36.52 dB, demonstrating an excellent impedance match ($Z_{\mathrm{in}} \approx 50\,\Omega$) within the passband.

2. **Quarter-Wave Open-Circuit Stub Filter:** The open coaxial stub ($L = 0.91$ m) behaved as a notch filter with maximum attenuation of 28.50 dB at 55.0 MHz. This closely matches the theoretical prediction $f_{0,\mathrm{calc}} = 54.40$ MHz (error $< 1.1\%$), where minor discrepancies stem from connector end-capacitance and velocity factor tolerances.

3. **RF Mixer:** Down-conversion and up-conversion were confirmed across $f_{\mathrm{RF}} = 13$–43 MHz under $f_{\mathrm{LO}} = 10$ MHz drive. The average conversion loss was 11.39 dB. Higher RF frequencies exhibited lower conversion losses (≈ 9.5 dB at 33–43 MHz).

4. **Power Splitter/Combiner:** The 2-way splitter demonstrated balanced power division across paths (3.52–3.57 dB loss vs. 3.0 dB ideal). The combiner test verified coherent voltage summation ($P_{\mathrm{comb}} = +3.05$ dBm vs. theoretical +3.48 dBm), exhibiting an internal insertion loss of 0.43 dB.